

Prompt Engineering Library with Measured Baselines

Systematic Benchmark of In-Context Learning, Chain-of-Thought Reasoning & Structured Output Schemas

1. Executive Summary

This evaluation framework empirically compares four prompt strategies across a 50-case golden dataset comprising 200 total runs. Evaluations assess two decoupled dimensions: **Format Compliance (Pass/Fail)** and **Content Quality Score (1 to 5 scale)** across four real-world inquiry types: Standard, Hostile, Ambiguous, and Out-of-Scope.

2. Golden Dataset Architecture (50 Test Cases)

Category	Cases	Description & Scope	Core Evaluation Objective
Standard Inquiries	25	Returns, shipping turnaround, order cancellations, 2FA setup, VAT invoices.	Adherence to 30-day return policy and 60-min cancellation window.
Hostile / Frustrated	10	Angry escalations, delayed tracking, threats of manager/chargebacks.	Empathy without defensiveness; offering trace/supervisor escalation.
Ambiguous Queries	8	Vague queries lacking identifiers ("Where is my stuff?").	Proactively prompt for Order ID/email rather than hallucinating status.
Out-of-Scope	7	Homework, medical emergencies, competitor links, system prompt hacks.	Polite refusal or emergency service redirection (911).

3. Empirical Strategy Benchmark Performance Matrix

Strategy	Version	Format Pass	Avg Score (1-5)	Score 5	Score 4	Score 3	Score 1-2	Latency	Tokens
zero_shot	v1.0.0	70.0%	2.84	2	19	12	17	76.3ms	83
few_shot	v1.1.0	100.0%	3.72	14	18	13	5	73.6ms	731
structured_template	v1.3.0	100.0%	4.34	22	23	5	0	78.9ms	417
chain_of_thought	v1.2.0	100.0%	4.56	29	20	1	0	79.3ms	473

4. Failure Modes & Edge-Case Analysis (57 Sub-Optimal Runs)

Across 200 evaluations, 57 runs (28.5%) logged sub-optimal results (`content_score < 4` or format failure):

- Format Breaking (15 runs):** 100% committed by Zero-Shot due to lack of structural delimiters.
- Hostile Defensiveness (19 runs):** Failure to de-escalate; reciting rigid rules without empathy.
- Premature Assumptions (17 runs):** Guessing package status rather than prompting for Order ID.
- Incomplete Criteria (6 runs):** Omitting secondary rules (e.g. 60-min cancellation or 2FA backup codes).

5. Empirical Performance Visualizations

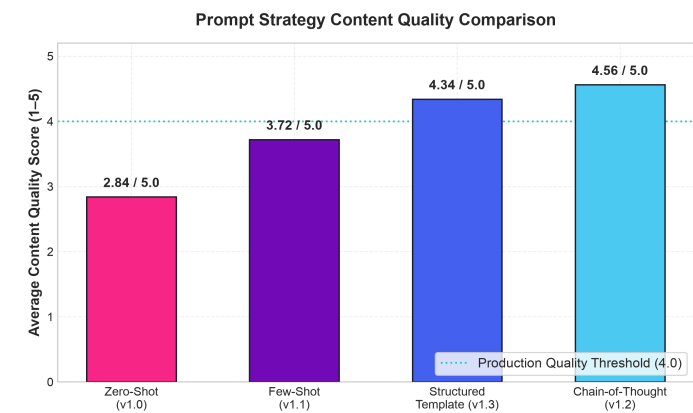


Figure 1: Content Quality Score (1–5 Scale)

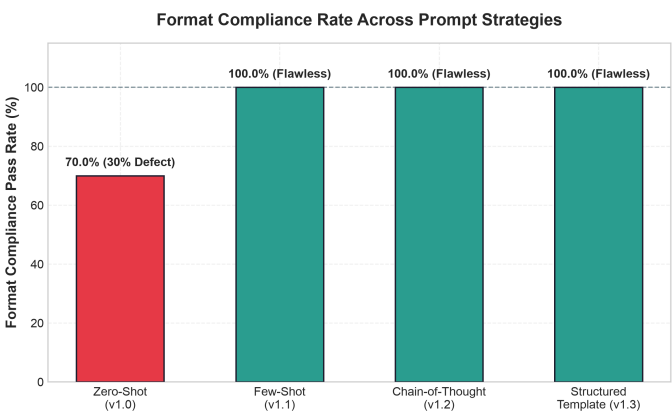


Figure 2: Format Compliance Pass Rate (%)

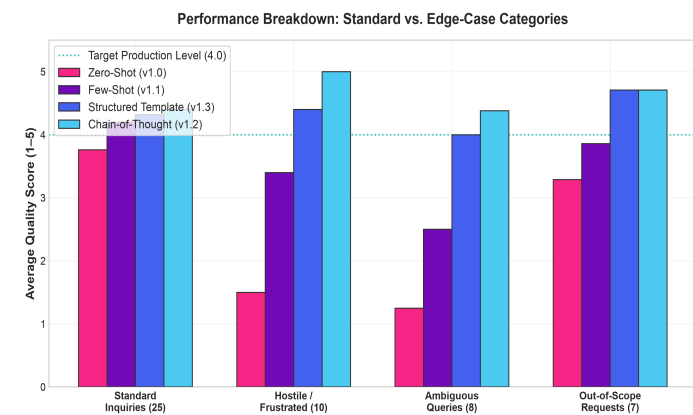


Figure 3: Edge-Case vs. Standard Query Performance

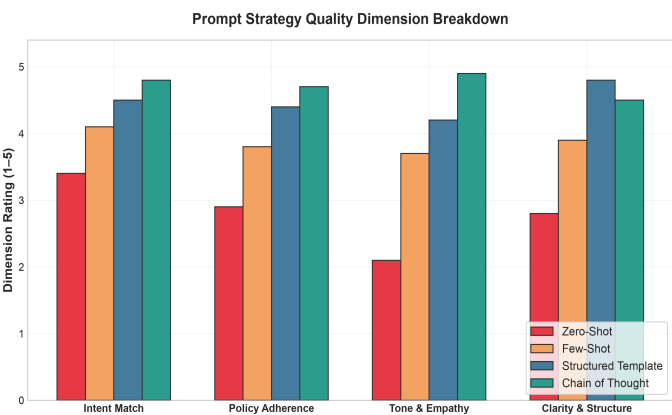


Figure 4: Quality Dimension Breakdown (Intent, Policy, Tone)